

Joint Probability Mass Function

Consider 2 D.R.V. X & Y , the proba of the values X & Y can take are captured by JPMF of X & Y .

$$P_{X,Y}(x,y) = P[X=x, Y=y]$$

$$\Rightarrow \text{Marginals} \rightarrow P_X(x) = \sum_{y \in \Omega_Y} P_{X,Y}(x,y)$$

$$P_Y(y) = \sum_{x \in \Omega_X} P_{X,Y}(x,y)$$

$$\Rightarrow Z = g(X,Y)$$

$$P_Z(z) = \sum_{\{x,y \mid g(x,y) \in z\}} P_{X,Y}(x,y)$$

Exercise

Waiting queue :

→ Number of clients in the queue follows the poisson dist. with parameter λ .

→ The proba that the client gets service is p .

Define the law & sample space of R.V. that counts number of satisfied clients.

⇒ X : number of clients in queue. $\frac{\lambda^k}{k!} e^{-\lambda}$
 $X \sim P(\lambda)$

$$\Omega_x = \{0, 1, \dots\}$$

$$P_x(k) = P[X=k] = \frac{\lambda^k}{k!} e^{-\lambda}$$

$$E[X] = \sum_{k=0}^{\infty} k \cdot \frac{e^{-\lambda} \lambda^k}{k!}$$

$$= 0 + \sum_{k=1}^{\infty} \frac{e^{-\lambda} \lambda^k}{(k-1)!}$$

$$= 0 + e^{-\lambda} \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}$$

$$= \underline{\underline{\lambda}}$$

X clients in queue

$$\Rightarrow \frac{e^{-\lambda} \lambda^x}{x!}$$

Y satisfied clients

$$Y \rightarrow \{0, 1, \dots, \infty\}$$

$\Rightarrow \gamma = i$ satisfied clients given there are $x = k$ clients in q .

$P(Y|x) \Rightarrow$ Binomial

$$\Rightarrow (p)^i (1-p)^{k-i} \cdot {}^k C_i$$

$$\Rightarrow p^i (1-p)^{k-i} \cdot C_i$$

$$\Rightarrow \boxed{P(Y=i | X=k) = \frac{P(X, Y)}{P(X)}}$$

$$P(X, Y) = (p^i \cdot (1-p)^{k-i} \cdot C_i) \cdot \frac{e^{-\lambda} \lambda^k}{k!}$$

Now to find the $P_X(x)$ using marginals.

$$P_Y(y=i) = \sum_k P_{X,Y}(x,y)$$

$$= \sum_{k=i}^{\infty} \left(p^i (1-p)^{k-i} \binom{k}{i} \right) \cdot \frac{e^{-\lambda} \lambda^k}{k!}$$

$$= p^i e^{-\lambda} \sum_{k=i}^{\infty} \frac{(1-p)^{k-i} \binom{k}{i} \lambda^k}{k!}$$

$$= p^i e^{-\lambda} \sum_{k=i}^{\infty} \lambda^k (1-p)^{k-i} \frac{\cancel{k!}}{(k-i)! i!} \times \frac{1}{\cancel{k!}}$$

$$= \frac{p^i e^{-\lambda}}{i!} \sum_{k=i}^{\infty} \frac{\lambda^k (1-p)^{k-i}}{(k-i)!}$$

$$= \frac{p^i e^{-\lambda} \lambda^i}{i!} \sum_{k=i}^{\infty} \frac{\lambda^{k-i} (1-p)^{k-i}}{(k-i)!}$$

$$= \frac{p^i e^{-\lambda} \lambda^i}{i!} \sum \frac{(\lambda(1-p))^{k-i}}{(k-i)!}$$

$$= \frac{p^i \cancel{e^{-\lambda}} \lambda^i}{i!} \cdot \frac{\cancel{e^{-\lambda}}}{e^{\lambda p}}$$

$$= \frac{(\lambda p)^i e^{-\lambda p}}{i!}$$

$$P[Y=i] = \frac{(\lambda p)^i e^{-\lambda p}}{i!}$$

$$\Sigma[Y] = \sum_{i=0}^{\infty} i \cdot \frac{(\lambda p)^i e^{-\lambda p}}{i!}$$

$$\Sigma [Y] = \sum_{i=0}^{\infty} i \cdot \frac{(\lambda p)^i e^{-\lambda p}}{i!}$$

$$= \sum_{i=1}^{\infty} \frac{(\lambda p)^i e^{-\lambda p}}{(i-1)!}$$

$$= e^{-\lambda p} \cdot \lambda p \cdot e^{\lambda p}$$

$$= \underline{\underline{\lambda p}}$$

⇒ Now for X

$$P_X [X=k] = \sum_{i=0}^k \left(p^i (1-p)^{k-i} {}^k C_i \right) \cdot \frac{e^{-\lambda} \lambda^k}{k!}$$

$$P_x [x = k] = \sum_{i=0}^k \left(p^i (1-p)^{k-i} \binom{k}{i} \right) \cdot \frac{e^{-\lambda} \lambda^k}{k!}$$

$$\Rightarrow \frac{e^{-\lambda} \lambda^k}{k!} \sum_{i=0}^k \cancel{p^i (1-p)^{k-i}} \binom{k}{i}$$

$$= \frac{e^{-\lambda} \lambda^k}{k!}$$

Exercise

We have an urn filled with n balls numbered from 1 to N .

We do k pullings without putting back the balls.

$X \rightarrow$ smallest t number obtained

i) $\Omega_x = ?$

ii) $P_x(i) = ?$

iii) Verify the normality axiom.

iv) Show that $E[X] = \sum_{i=1}^{n-k+1} P[X \geq i]$

i) $\Omega_x = \{1, 2, \dots, n-k+1\}$

ii) total counts = ${}^n C_k$

$$P_x(x=i) = \frac{1 \cdot {}^{n-i} C_{k-1}}{{}^n C_k} = \frac{{}^{n-i} C_{k-1}}{{}^n C_k}$$

$$= \frac{(n-i)!}{(k-1)!(n-k+1)!} \cdot \frac{k!(n-k)!}{n!}$$

$$\frac{(n-i)!}{n!} \cdot k \cdot \frac{1}{n-k+1}$$

$$= \frac{(n-i)!}{n!} \cdot \frac{k}{n-k+1} \Rightarrow \frac{{}^{n-i}C_{k-1}}{{}^nC_k}$$

$$\text{iii)} \sum P_x [x=i] = \sum_{i=1}^{n-k+1} \frac{{}^{n-i}C_{k-1}}{{}^nC_k}$$

$$\frac{1}{{}^nC_k} \sum_{i=1}^{n-k+1} {}^{n-i}C_{k-1}$$

$${}^nC_k = {}^{n-1}C_{k-1} + {}^{n-1}C_k$$

$$\frac{1}{{}^nC_k} \left[{}^{n-1}C_{k-1} + {}^{n-2}C_{k-1} + \dots + {}^{k-1}C_{k-1} \right]$$

$$\frac{1}{{}^nC_n} \left[\underbrace{{}^{n-1}C_{k-1} + {}^{n-2}C_{k-1} + \dots + {}^{k-1}C_{k-1}}_{{}^nC_n} \right]$$

$$= 1$$

iv) $E[X] = \sum_{i=1}^{n-k+1} P[X \geq i]$

$$E[X] = \sum_{i=1}^{n-k+1} i \cdot \frac{{}^{n-i}C_{k-1}}{{}^nC_n}$$

$P[X \geq i] \text{ ① } \rightarrow P[X=1] + P[X=2] + \dots + \dots$
 $\text{② } \quad \quad \quad P[X=2] + \dots + \dots$
 $\text{③ } \quad \quad \quad + P[X=3] + \dots + \dots$

$$\sum_{i=1}^{n-k+1} P[X \geq i] = 1 \cdot P[X=1] + 2 \cdot P[X=2] \dots$$

$$= \sum i \cdot P[X=i]$$

Hence proved

Continuous Random Variable

$\Rightarrow$ PDF $\rightarrow$ Probability density function.

$\rightarrow$ A R.D.V is called continuous if there is a non-negative function $f_X(x)$ called the probability density function such that $P[X \in B]$

$$P[x \in B] = \int_{x \in B} f_x(x) dx$$

In particular

$$\bullet P[a \leq x \leq b] = \int_a^b f_x(x) dx$$

$$\bullet P_x[x = k] = 0$$

Expectation

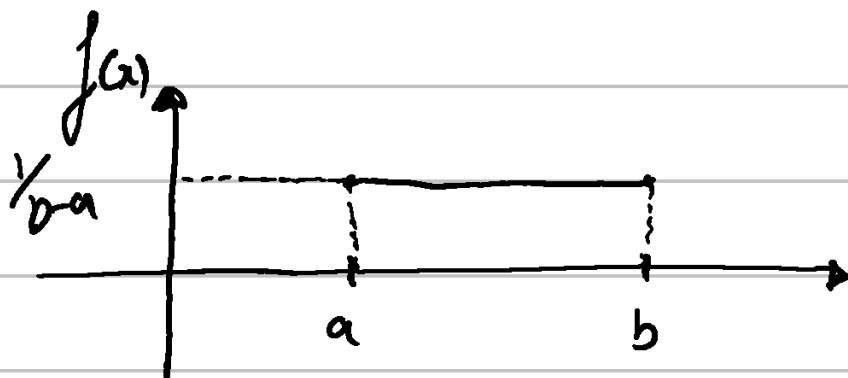
$$E[x] = \int_{x \in \Omega_x} x \cdot g_x(x) dx$$

$$E[g(x)] = \int_x g(x) \cdot f(x) dx$$

Famous PDF

① Uniform r.d.v.

$$x \sim U[a, b]$$



$$E[x] = \int_a^b \frac{x}{b-a} dx$$

$$= \left[\frac{x^2}{2(b-a)} \right]_a^b = \frac{b^2 - a^2}{2(b-a)}$$

$$E[x] = \frac{a+b}{2}$$

$$\text{Var}[x] = 0$$

